



Spring School
AI For Impact

AI Email Assistant with Personalized Style Learning

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Introduction & Objectives

The Problem: Professionals spend hours each week writing email replies, repetitive, time-consuming, and mentally draining. Existing AI tools can generate responses, but they sound generic, robotic, and nothing like you.

Objectives:

- **Personalization:** Extract 17 linguistic metrics from your sent emails to build a writing profile that is uniquely yours.
- **Security Multi-layer defense:** injection guard, output validation, AES-256-GCM encryption & HMAC-SHA256 signing.
- **Human-in-the-Loop:** AI drafts, you decide. Nothing is ever sent automatically.

Main Contribution

- **Style Engine:** Learns your writing style from 25 sent emails across 17 metrics.
- **Adaptive Drafting:** Groq AI replies in your tone using 3 real samples of your writing.
- **Security Pipeline:** 5-layer protection: injection guard → validation → encryption → signing → audit log.
- **Privacy-First:** No raw personal data ever stored.

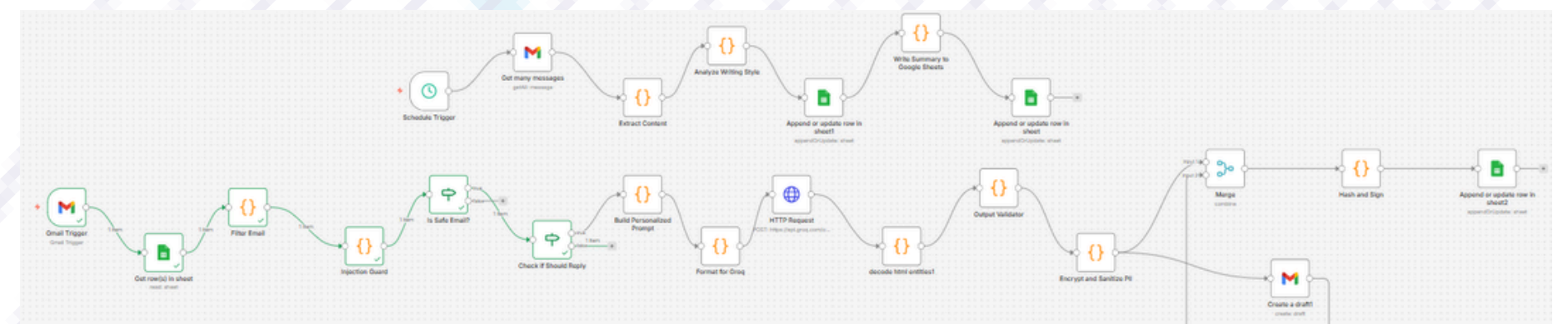
Conclusion & Perspectives

- **Conclusion:** A secure AI assistant that drafts emails in your voice, not a generic one.
- **Perspectives:** RAG for company-specific context - Feedback loop to improve from your edits - Multi-user team support.

Bibliography

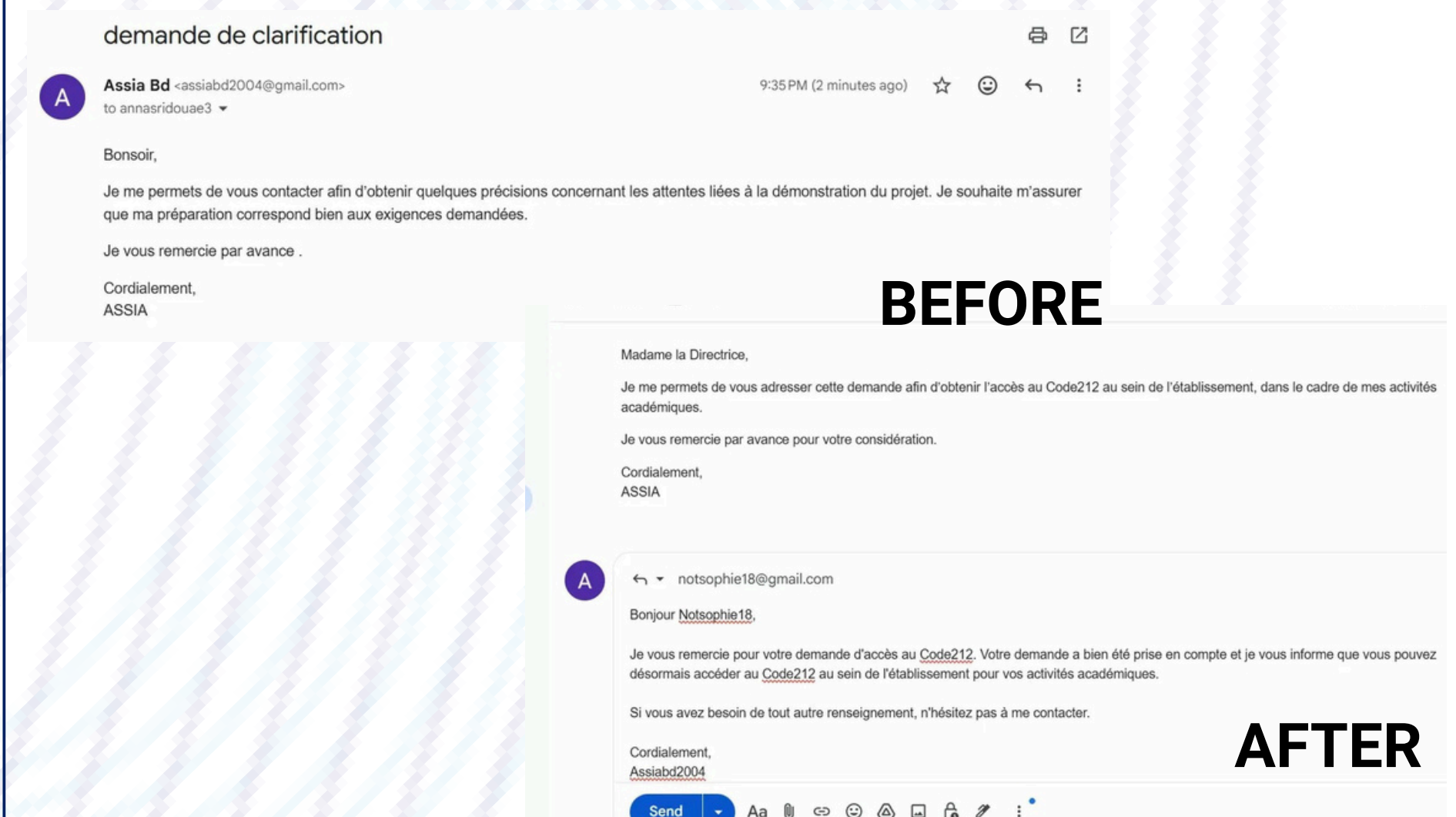
<https://docs.n8n.io/>; <https://console.groq.com/docs/api-reference>; <https://console.groq.com/docs/overview>

Methodology, Analysis & Results



Workflow 1 – Style Learner (runs on schedule) Fetches 25 sent emails → extracts 17 style metrics → saves profile to Google Sheets.

Workflow 2 – Draft Generator (triggers on new email) Filters & guards → loads style profile → Groq AI → validates → encrypts → saves to Gmail drafts.



Performance Metrics:

- 70B parameter AI model
- Draft generated in < 2 seconds
- Any language, auto-detected
- Style pre-calculated, not recomputed per email
- 25 sent emails analyzed per profile